

Sharing & Unit Conversions

Use ratios to share a quantity fairly, and convert between units of length, area, volume and temperature.

Sharing a quantity in a ratio

The idea: To divide a quantity x in the ratio $m : n$, split it into $(m + n)$ equal groups. Each group is $x \div (m + n)$.

$$\text{First part} = m / (m + n) \times x \quad | \quad \text{Second part} = n / (m + n) \times x$$

Example: share 42 in the ratio 4 : 3. Groups = $4 + 3 = 7$, each group = $42 \div 7 = 6$, so the parts are $4 \times 6 = 24$ and $3 \times 6 = 18$.

Handy unit conversions

Quantity	Conversion
Length	1 metre = 3.281 feet
Area	1 hectare = 10,000 m ² ; 1 acre = 43,560 ft ²
Volume	1 litre = 1000 mL; 1 mL = 1 cm ³
Temperature	$^{\circ}\text{F} = (9/5 \times ^{\circ}\text{C}) + 32$; $^{\circ}\text{C} = 5/9 \times (^{\circ}\text{F} - 32)$

Check yourself: after sharing, the parts should add back to the whole ($24 + 18 = 42$).

Prerequisite refresher: $m / (m + n)$ means "m out of every (m + n) parts". The two fractions add to 1.

Where marks slip away

Trap 1 — Use the total number of parts. For ratio 4 : 3, divide by 7 ($= 4 + 3$), not by 4 or by 3.

Trap 2 — Match part to ratio number. The 4-share gets 4 groups, the 3-share gets 3 groups.

Trap 3 — Convert before comparing. A weight-to-price ratio of 1 kg : ₹800 is really 1000 g : 800, depending on the unit you fix.

Trap 4 — Temperature isn't a simple ratio. Use the formula $^{\circ}\text{F} = 9/5 \times ^{\circ}\text{C} + 32$; you cannot just multiply by a single factor.

Slip 5 — Add the parts back to check they total the original quantity.

Model answers — write them like this

Example A. Share 42 counters in the ratio 4 : 3.

Step 1: Total parts = $4 + 3 = 7$; each part = $42 \div 7 = 6$.

Step 2: Shares = $4 \times 6 = 24$ and $3 \times 6 = 18$.

$\therefore 24$ and 18

Example B. A mortar mix of 40 kg has sand : cement = 3 : 1. Find each weight.

Step 1: Total parts = $3 + 1 = 4$; each part = $40 \div 4 = 10$ kg.

Step 2: Sand = $3 \times 10 = 30$ kg; cement = $1 \times 10 = 10$ kg.

$\therefore 30$ kg sand, 10 kg cement

Example C. Convert 25°C to Fahrenheit.

Step 1: $^\circ\text{F} = (9/5 \times ^\circ\text{C}) + 32 = (9/5 \times 25) + 32$.

Step 2: $= 45 + 32 = 77$.

$\therefore 25^\circ\text{C} = 77^\circ\text{F}$

Example D. The masses of equal volumes of gold and water are in the ratio 37 : 2. If 1 litre of water is 1 kg, find the mass of 1 litre of gold.

Step 1: gold : water = 37 : 2 and water = 1 kg, so 1 kg matches the "2" part.

Step 2: One part = $1 \div 2 = 0.5$ kg, so gold = $37 \times 0.5 = 18.5$ kg.

$\therefore 18.5$ kg

Next: [try the worksheet](#), then [check the answer key](#).